

Limits on Understanding That Emerge from Scale

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Question. Suppose an experiment can observe physical structure only up to a length R . Should we expect related structure at lengths larger than R ? This note separates two ways to answer. A probability model can assign a chance to larger scales. A fluid equation can explain how energy reaches them. Neither approach allows a finite observation, by itself, to prove continuation to arbitrarily large scales.

Running example. To keep the discussion concrete, consider a two-dimensional flow forced at length $L_f = 1$ m. The forcing supplies energy at rate $\varepsilon = 10^{-3} \text{ m}^2 \text{ s}^{-3}$, and our experiment can detect coherent structures only up to $R = 8$ m. We will ask what, if anything, can be said about structures larger than 8 m.

1. A probability law for further scales

Scales as a point process

Suppose that every structure we care about, such as an eddy, wave, or instability, has a characteristic length L . We record only that length. The collection of structures is then represented by a random set of points on the positive L -axis, starting at a chosen minimum length $L_{\min}$.

It is natural to measure this axis logarithmically: going from 1 to 2 metres is the same relative change as going from 100 to 200 metres. We now make a specific modeling assumption. Counts in nonoverlapping scale intervals are independent, and the expected number in a short interval near L is

$$d\mu(L) = \eta(L) \frac{dL}{L}, \quad \eta(L) \geq 0.$$

These two statements define a Poisson point process. The function $\eta(L)$ is the expected density of structures per unit of log scale. A structure is “relevant” only if it interacts with the phenomenon being studied. If one instead measures relative d -dimensional volume, the extra factor d can be absorbed into η and does not change the argument.

The number $N(R, S)$ of processes with scales in $(R, S]$ is Poisson with mean

$$\mu(R, S) = \int_R^S \eta(L) \frac{dL}{L}.$$

The chance of seeing at least one such structure is therefore

$$\mathbb{P}\{N(R, S) \geq 1\} = 1 - \exp\left[-\int_R^S \eta(L) \frac{dL}{L}\right]. \quad (1)$$

Now let the upper limit S grow without bound. The model gives the exact test

$$\boxed{\text{unbounded process scales almost surely}} \iff \boxed{\int_R^\infty \eta(L) \frac{dL}{L} = \infty}. \quad (2)$$

Here “almost surely” means with probability one under the model.

Coin-flip intuition

First suppose $\eta(L) = \eta_0 > 0$ at every scale. Each doubling interval $(2^j R, 2^{j+1} R]$ has the same hit probability

$$p = 1 - \exp(-\eta_0 \log 2) = 1 - 2^{-\eta_0}.$$

The intervals are independent, so the probability of missing the first m is $2^{-m\eta_0} \rightarrow 0$. This is like repeating a coin toss with a fixed positive chance of heads: eventually a head appears, and in fact heads appear infinitely often. Likewise, structures occur in infinitely many doubling intervals with probability one. Arratia calls this a scale-invariant Poisson process [1].

In the running example, take $\eta_0 = 1$. Each interval $(8, 16]$, $(16, 32]$, $(32, 64]$, and $(64, 128]$ metres then has probability $1/2$ of containing a relevant structure. The probability of finding at least one by 128 m is

$$1 - \left(\frac{1}{2}\right)^4 = 0.9375.$$

If the same constant intensity continues through infinitely many doubling intervals, structures occur at arbitrarily large scales with probability one.

A decaying tail

The answer changes if large structures become sufficiently rare. For example, if $\eta(L) = \eta_0(L_{\min}/L)^a$ with $a > 0$, then

$$\int_R^\infty \eta(L) \frac{dL}{L} = \frac{\eta_0}{a} \left(\frac{L_{\min}}{R} \right)^a < \infty.$$

then the total expected number beyond R is finite. The probability that no relevant structure exists beyond R is therefore

$$\exp \left[-\frac{\eta_0}{a} \left(\frac{L_{\min}}{R} \right)^a \right]$$

For a numerical contrast, define two models that agree throughout the observed range:

$$\eta_A(L) = 1, \quad \eta_B(L) = \begin{cases} 1, & L \leq 8 \text{ m}, \\ (8 \text{ m}/L)^2, & L > 8 \text{ m}. \end{cases}$$

Model A predicts unbounded scales almost surely. For model B ,

$$\int_{8 \text{ m}}^\infty \left(\frac{8 \text{ m}}{L} \right)^2 \frac{dL}{L} = \frac{1}{2}, \quad \mathbb{P}\{\text{no structure beyond } 8 \text{ m}\} = e^{-1/2} \approx 0.607.$$

Why finite data cannot decide

Models A and B fit every possible observation made at $L \leq 8 \text{ m}$, yet they give opposite answers about unlimited continuation. The answer is determined by the assumed tail of η , which the finite experiment cannot see.

2. What Navier–Stokes can supply

The Poisson model describes how many scales occur, but it does not explain what creates them. Two-dimensional fluid flow provides a possible mechanism: an inverse cascade can carry energy from the forcing scale toward larger lengths. Consider the randomly forced, damped Navier–Stokes equations on a periodic square $\mathbb{T}_\lambda^2$ of size λ :

$$du + [(u \cdot \nabla)u + \nabla p] dt = [\nu \Delta u - \alpha(-\Delta)^{-2\gamma} u] dt + dW^\lambda, \quad \nabla \cdot u = 0. \quad (3)$$

Here u is velocity, p is pressure, ν is viscosity, α is large-scale drag, and W^λ is the random forcing. The forcing injects mean energy $\varepsilon > 0$ per unit time and area near one fixed length.

The inverse-flux law

Bedrossian, Coti Zelati, Punshon-Smith, and Weber proved an inverse-flux law for stationary solutions as the box grows and viscosity and drag vanish [2]. Among their assumptions are

$$\nu \mathbb{E} \|\omega\|_2^2 \rightarrow 0, \quad \alpha \mathbb{E} \|(-\Delta)^{-\gamma} \omega\|_2^2 \rightarrow 0. \quad (4)$$

where $\omega = \text{curl } u$ is vorticity. The powers -2γ and $-\gamma$ are consistent: a quadratic energy norm splits the operator equally between its two factors, as in Ref. [2].

To measure transfer across a length ℓ , compare velocities at two points separated by ℓ in direction n : $\delta_{\ell n} u(x) = u(x + \ell n) - u(x)$. Average the resulting cubic quantity over all positions x and directions n :

$$\mathcal{F}_\ell(u) = \int_{S^1} \int_{\mathbb{T}_\lambda^2} |\delta_{\ell n} u|^2 (\delta_{\ell n} u \cdot n) dx dn.$$

The theorem gives an upper inertial-range length $\ell_\alpha \rightarrow \infty$ for which

$$\mathbb{E} \mathcal{F}_\ell(u) \sim 2\varepsilon \ell, \quad 1 \ll \ell \ll \ell_\alpha. \quad (5)$$

The positive sign says that energy is moving toward larger lengths. Thus, for any chosen finite R , one can take a large enough box and weak enough drag to obtain active inverse transfer at some $\ell \geq R$. Notice the quantifiers: the system may change when R changes. This does not yet describe one flow whose active scale grows forever.

Numerical flux at 8 metres

For the running example, suppose 8 m lies inside the inertial range. Equation (5) then gives

$$\mathbb{E}\mathcal{F}_{8\text{ m}} \approx 2(10^{-3}\text{ m}^2\text{ s}^{-3})(8\text{ m}) = 0.016\text{ m}^3\text{ s}^{-3}.$$

This positive number says that energy is crossing the 8 m scale in the direction of still larger structures. It supplies a physical mechanism for continuation beyond the measurement limit. It does not prove that the scale of one particular flow grows without bound.

The role of ergodicity

There is also an almost-sure statement at each fixed set of parameters. If the chosen stationary measure is ergodic, Birkhoff's theorem gives

$$\frac{1}{T} \int_0^T \mathcal{F}_\ell(u(t)) dt \longrightarrow \mathbb{E}\mathcal{F}_\ell(u) > 0. \quad (6)$$

So the long-time average flux is positive for almost every realization. This is a stationary result for an allowed scale in a fixed box. It does not remove the box's largest possible length.

3. The stronger theorem still wanted

The stronger goal concerns one flow on the whole plane $\mathbb{R}^2$. A spatially homogeneous flow on an infinite plane generally has infinite total energy, so the right object is a homogeneous statistical solution with finite energy *per unit area*, not a finite-energy $L^2(\mathbb{R}^2)$ solution.

Defining the large scale

We must also define what the flow's "large scale" means. Let $E(k, t)$ be its energy spectrum, where small wave number k means large physical length. Fix a fraction $\theta \in (0, 1)$ and define

$$k_\theta(t) = \inf \left\{ K > 0 : \int_0^K E(k, t) dk \geq \theta \int_0^\infty E(k, t) dk \right\}, \quad L(t) = k_\theta(t)^{-1}.$$

Thus $k_\theta(t)$ is the wave number below which the chosen fraction of energy lies, and $L(t) = 1/k_\theta(t)$ is its associated length. Energy moving toward smaller k makes $L(t)$ larger. The desired theorem is

$$\mathbb{P}\{L(t) \rightarrow \infty\} = 1. \quad (7)$$

A conditional growth law

What growth rate might we expect? If we additionally assume Kraichnan's inverse-cascade spectrum [3], with forcing length L_f ,

$$E(k) = C_I \varepsilon^{2/3} k^{-5/3}, \quad L(t)^{-1} \leq k \leq L_f^{-1}.$$

then the energy accumulated by time t is approximately the integral of this spectrum. Setting that integral equal to εt gives

$$\varepsilon t = \frac{3C_I}{2} \varepsilon^{2/3} (L(t)^{2/3} - L_f^{2/3}),$$

so

$$L(t) = \left[L_f^{2/3} + \frac{2}{3C_I} \varepsilon^{1/3} t \right]^{3/2} \sim \left(\frac{2}{3C_I} \right)^{3/2} \varepsilon^{1/2} t^{3/2}. \quad (8)$$

For a simple numerical illustration, set $C_I = 1$. This choice makes the arithmetic transparent; it is not an empirical calibration. Using the values from the running example, (8) becomes

$$L(t) = 1\text{ m} \left(1 + \frac{t}{15\text{ s}} \right)^{3/2}.$$

The predicted scale reaches the observation limit after 45 seconds,

$$L(45\text{ s}) = 1\text{ m} (1 + 3)^{3/2} = 8\text{ m},$$

and reaches 27 m after 120 seconds. These numbers illustrate what the $t^{3/2}$ law says; they are conditional on the assumed spectrum.

What remains open

This calculation predicts $L(t) \propto t^{3/2}$, but it is not a theorem derived from (3): it assumes the spectrum that produces the answer. The lower spectral edge used in the calculation is proportional to the quantile length above, up to a constant that does not change the exponent.

Finally, this growing state is not stationary, so the Birkhoff argument in (6) cannot prove (7) or (8). Drag, finite forcing time, or a container may stop the growth. The open problem is to prove from Navier–Stokes that the infrared scale of one unbounded flow grows almost surely, under explicit assumptions, without first assuming the $k^{-5/3}$ law.

References

- [1] R. Arratia, “On the Central Role of the Scale Invariant Poisson Processes on $(0, \infty)$,” DIMACS Series **41** (1998), [arXiv:1611.05572](#).
- [2] J. Bedrossian, M. Coti Zelati, S. Punshon-Smith, and F. Weber, “Sufficient Conditions for Dual Cascade Flux Laws in the Stochastic 2d Navier–Stokes Equations,” *Arch. Ration. Mech. Anal.* **237** (2020), [103–145](#).
- [3] R. H. Kraichnan, “Inertial Ranges in Two-Dimensional Turbulence,” *Phys. Fluids* **10** (1967), [1417–1423](#).